# PromoFinder: finding promo clips in microdrama screenplays

Kuku AI Generalist Assignment · Problem 1 (Find the Promos) · Om Prakash Gupta · Roll No. 230104067

## 1. WHY THIS PROBLEM, AND SCOPE

I picked Problem 1 because it is one editor task with a checkable output: exact timestamps plus a hook, a cliffhanger and a reason. The tool only **finds** promos for an editor to approve. It does no cutting, music or overlays, and it is a shortlist, not a verdict.

## 2. ASSUMPTIONS (THE BRIEF WAS SILENT)

- Script timestamps equal the final edit. Promo length is the sum of its segments, 20–60 s, best at 25–45 s.
- The "first 3 s" hook is judged on dialogue, action and sound in the first 3 s. Camera lines are context only.
- A promo has at most 3 segments. Cross-episode clips join an episode ending to the next episode's opening.
- Spoilers are flagged, not blocked. English only. US audience, vertical short-form.
- No ground-truth labels exist, so quality comes from rules, a second AI pass and a small human review.

## 3. HOW IT WORKS: WHAT AI DOES, WHAT CODE DOES

| Step | Who | What happens |
|---|---|---|
| 1 Parse | Code | Regex reads every timed block (dialogue, action, sound, camera). Camera text is merged into the line it overlaps. Unparsed lines are counted: **0 of 3,727 blocks** failed across 50 files. |
| 2 Tag beats | AI | One call per episode scores each beat 0–10 for hook, cliffhanger and context needed, plus a trope tag. The model returns **beat IDs only**, never timestamps, so it cannot invent a time. |
| 3 Build clips | Code | Start on a strong hook beat, end on a strong cliff beat, snap to line boundaries (never mid-line), enforce 20–60 s, drop near-duplicates, keep the top 8 per episode. Also builds 2-segment stitches and cross-episode pairs. |
| 4 Judge | AI | Sees **only the clip text** (like a first-time viewer): hook line, cliffhanger line, one-line reason, standalone, spoiler risk. |
| 5 Rank | Code | `0.30 hook3s + 0.30 cliff + 0.20 standalone + 0.15 emotion + 0.05 length-fit − spoiler penalty`, plus a diversity penalty so the top list is not one trope or one scene. |
| 6 QC | Code + AI + human | Rule checks, a skeptical second-pass critic prompt, and a manual review sample (see section 5). |

The model is swapped with three environment variables (`LLM_PROVIDER`, `LLM_MODEL`, `LLM_API_KEY`), so a stronger model can be connected without code changes. Replies are cached, so re-runs cost nothing. Runtime model for this run: **Gemini 3.1 Flash Lite** (free tier).

## 4. RESULT ON THE FULL 50-EPISODE PACK

**180 promos**: 65 single-stretch, 103 stitched, 12 cross-episode; average length 29.5 s. Ranked list and the top 5 to post first with reasons are in the raw outputs and on the Results page. My top picks:

| Rank | Where (episode, mm:ss) | Type · tag | Why I would post it (my reading of the model's rationale) |
|---|---|---|---|
| 1 | Ep 31 02:13–02:40 | single · threat | Opens on a burning-book image, ends on a hot-poker close-up. Strong and self-explanatory, but both ends are visual/sound text, so check on video. |
| 2 | Ep 23 01:19–01:30 + 01:58–02:11 | stitched · reveal | Opens on a poisoning accusation, ends on an unresolved door-lever moment. |
| 3 | Ep 3 01:24–01:36 + 02:13–02:25 | stitched · rescue | Action opener joined to an emotional, unanswered proposal ending. |
| 4 | Ep 36 02:16–02:28 + 03:04–03:20 | stitched · threat | Opens on an execution threat with a deadline. |
| 9 | Ep 1 02:04–02:17 + Ep 2 00:00–00:12 | cross-episode · betrayal | Early-funnel "previously on": the altar humiliation, continued into episode 2. Spoiler risk low. |

I would post early-episode, low-spoiler clips first (they funnel viewers into episode 1) and avoid the 15 clips the judge marked high spoiler risk until later in a show's run.

## 5. HOW I KNOW IT IS GOOD, AND WHERE IT FAILS

### Checks run

- **Rules (code):** 155 of 180 (86%) pass all. Flags: 21 weak cliffhanger (last beat scores under 6), 4 with no dialogue/action/sound in the first 3 s.
- **Second-pass critic:** flags 118 as confusing to a new viewer and 32 as resolving the tension; in 8 it disagrees with the judge. It is harsh, so I treat it as a ranking hint, not a veto.
- **Unit tests:** 39 pass (parser, snapping, windows, ranking, QC).

### Where it picks badly (from the real run)

- **Sound/visual "cliffhangers":** some clips end on a sound-effect swell or a camera description. The model counts these as unresolved moments; text cannot confirm they land on screen.
- **Missed episodes:** 5 episodes return no promo. Eps 6, 8, 17 are quiet scenes (best scores ≤4), which looks right. Ep 7 has no beat scoring 7+ as a cliffhanger. Ep 22 has hook 8 and cliff

> **Not done yet:** human precision@5/10/20 is **not measured**. The labelling file and the code that reports it are built; I have not reviewed clips on video, so I make no accuracy claim.

10, yet no valid clip was built: the clip builder is stricter than the tagger scores suggest.

- **Stitches may feel disjoint:** 103 of 180 are stitched and none has been watched. Cross-episode joins are only ep N end + ep N+1 start.
- **Text only:** acting, framing and audio are invisible.

## 6. SCALE, COST AND THE HUMAN IN THE LOOP

About 3 AI calls per episode (tag, judge, critic) plus 2 per consecutive pair for cross-episode clips: roughly 250 calls per 50-episode show, well under 1M tokens. One episode took ~18 s end to end in a test; the full run was slower only because the free tier rate-limited me (the adapter now backs off and resumes from cache). Cost: free tier here; I have not computed paid pricing and would do that from the current price sheet. Work is per episode, so it scales linearly across shows, can run in parallel, and can use batch APIs (typically about half price). An editor approves the top 10 per show; rejections become labels to tune the ranking weights and prompt examples.

## 7. AI TOOLS USED, AND WHERE THEY GOT IT WRONG

**Claude Code** drafted the code and tests; **Gemini 3.1 Flash Lite** is the runtime model. **Where AI got it wrong:** the assistant defaulted to `gemini-2.5-flash`, which Google had retired for new users, so the first real call failed with a 404. The replacement it suggested (`gemini-3.8-flash`) worked but took about 3 minutes per call, which was impractical, so I listed the available models and switched to the lite variant (~18 s per episode). Separately, Gemini scored a sound-effect swell as a strong cliffhanger (section 5), which is why a human check on video stays in the loop.